

Figure 1: The workstation with the forceps close to the retina and the left jaw nearer (pose set by hand in the simulation mode; all screenshots are of the running application). Left column, read top to bottom from coarse to fine: **1** estimated clearance of the nearer tip with status word and symbol (Q1); **2** accuracy note; **3** nearest tip and one depth bar per tip (Q2); **4** approach trend of the last 10 s with motion word and rate (Q3); **5** instrument context (jaws, insertion, angle); **6** proximity scope, a magnified view of the retina below the jaw midpoint with a drop line from the nearer tip. 3D view: **7** scene alarm (frame and banner); **8** instrument shadow, drop line, landing reticle and distance label (inset: dashed region, magnified; the short instrument on the left is the light pipe); **9** view presets; **10** model assumption (spherical eye, $R = 12$ mm); **11** data source and connection.

1 Visualisation: the depth-guidance workstation

The tracker delivers the 3D positions of both forceps jaw tips for each microscope frame it can reconstruct (about 15 Hz live). The workstation (`services/viz`, React and three.js in the browser, fed over a WebSocket) turns these positions into one quantity, the *clearance* of the instrument above the retina. It is a proof of concept for training on the Eyesi simulator, not a medical device. Its readers are the trainee, who glances at it between looks into the oculars, and an instructor watching the same screen live. The design follows one rule: the most important answer must be readable first and at a glance, with detail behind it. We reduced the task to three questions, which loosely map onto the perception, comprehension and projection levels of situation awareness [2]: **Q1** How close is the instrument to the retina? **Q2** Which of the two tips is closer? **Q3** Is it getting closer?

1.1 Layout and encoding

Figure 1 shows the screen. The left column answers the three questions in reading order; the 3D view on the right provides spatial context and repeats the alarm, but never holds the only copy of a number.

Q1: one number. The clearance of each tip is its radial distance to a sphere of radius 12 mm, $d = \max(0,\ 12\,\text{mm} - \|\mathbf{p}_{\text{tip}}\|)$ with $\mathbf{p}_{\text{tip}}$ measured from the eye centre, and the large number shows

the minimum over both tips. It has one decimal and is rounded *down*. Two decimals would claim a resolution about 60 times finer than the tracker's median height error of 0.63 mm, while a reported value should carry no more digits than its uncertainty justifies [4]; whole millimetres would hide useful distinctions around the 0.5 mm threshold. Rounding down keeps the number from ever exceeding the estimate and makes it agree with the instantaneous level at a threshold (0.46 mm reads "0.4" next to Critical).

Status levels. Three provisional training levels, *Clear* ($\geq$ 2 mm), *Near retina* (0.5–2 mm) and *Critical proximity* ($<$ 0.5 mm), are each coded three times: as colour, as word and as symbol (●, ▲, ■). The level can therefore be read without colour. The palette (teal, yellow, red) moves "safe" from green to teal, away from the red–green confusion axis [8]; yellow also differs in lightness, while teal and red differ mainly in hue, which is why word and symbol are needed. Under simulated full deuteranopia [6], the colour difference between safe and danger rises from $\Delta E_{00} = 7.8$ (earlier green/amber/red palette) to 35.3. In simulation mode, as in Figure 1, the accuracy note (2) states that the geometry is exact. In live mode it summarises the simulator recordings for the level currently shown (Figure 2b): for Critical and Near, the 10th–90th percentile of the logged midpoint clearance among measured frames at that displayed level; for Clear, the median bias.

Q2 and Q3. The two depth bars place each tip on a square-root scale from 0 to 10 mm, so the decision-relevant first 2 mm take 45 % of the bar instead of 20 % on a linear scale. "Nearest tip: L / R / L ≈ R" names the nearer jaw. A tie band (enter below 0.10 mm, leave above 0.15 mm) absorbs noise between the tips; adding a 500 ms dwell cut label changes from 71 to 10 per minute in the replay. The approach trend uses the same scale and threshold lines as the bars, so a level reads the same in both. Because the slope on a square-root axis is not the velocity, the rate is printed as a number next to a motion word (*Approaching*, *Holding steady*, *Retracting*). An earlier time-to-contact countdown was removed: it divided a biased distance by a noisy velocity and looked like a measurement.

3D cues. The overview alone cannot show sub-millimetre gaps. Three cues make depth visible in 3D (inset and scope in Figure 1): a shadow cast from the light pipe (its position is fixed by the calibration), a strong cue for motion in depth [5]; a drop line from the nearer tip to the retina; and a reticle where that line meets the retina. Like the "Nearest tip" label, the drop line follows the stabilised nearer-tip selection (the midpoint for a tie), while the number stays the instantaneous minimum. The proximity scope (6) is a second, linked view [7] of the retina directly below the jaw midpoint. It turns on below 5 mm and keeps a fixed place in the layout, so the screen does not rearrange at the moment of highest risk.

1.2 Alert logic

A raw threshold applied frame by frame makes the status flicker. Measurement noise makes it chatter at each threshold, and in the offline replay 26.5 % of frames lack a reconstruction (including visibility and detection failures), so without a grace period every gap would end an alarm. The workstation therefore separates *availability* (is there a current measurement?) from the *distance level* (how close is it?) and uses asymmetric rules (Figure 3). Escalation is immediate: a warning must never arrive late because of filtering. De-escalation requires the same safer candidate level, beyond a hysteresis margin of 0.08 mm (about twice the estimated per-sample noise), for 250 ms of uninterrupted measurements; a change of candidate or a gap restarts this dwell. While a level is retained this way, the status reads, e.g., "Critical proximity · clearing". Missing data is never shown as safe. A missing or incomplete frame, or 200 ms without any new frame, holds the last values, dimmed and labelled "Signal gap". Once 400 ms have passed since the last valid measurement, the level is replaced by "No measurement" and the last pose is shown as a faded ghost. A source whose timestamp stops advancing for 1.5 s is marked stale. Figure 4 shows the real pipeline code processing a synthetic feed.

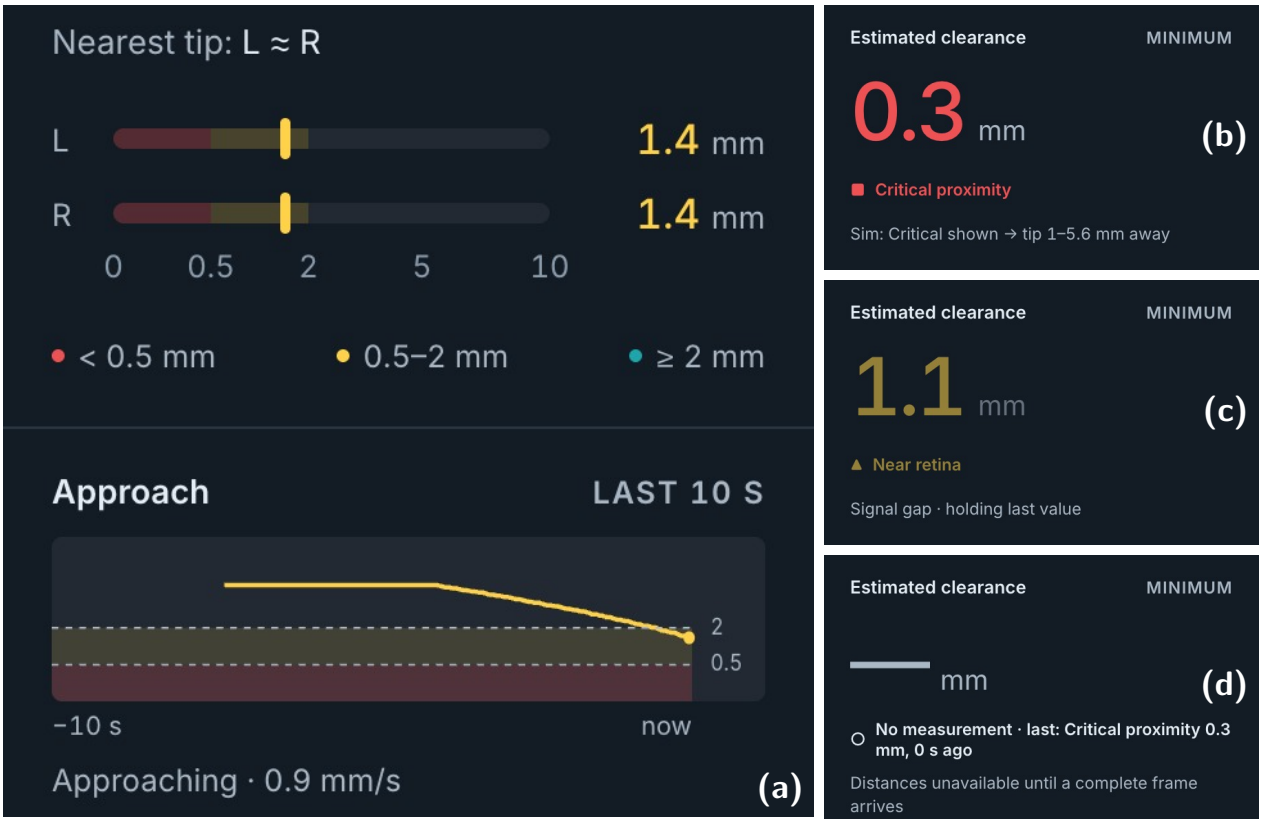


Figure 2: Left-column details, enlarged. (a) Nearest tip, depth bars, colour key and approach trend during a scripted approach in simulation mode (Q2, Q3); the whole trend line takes the colour of the current level. (b)–(d) The clearance card in live mode, driven by synthetic poses through a mocked WebSocket. In (b) the accuracy note reports that, when Critical was shown on the simulator recordings, the logged midpoint clearance was mostly 1–5.6 mm (10th–90th percentile). (c) A short signal gap: the last value is held and dimmed. (d) Once 400 ms have passed since the last measurement, the number disappears and the last reading is shown as history only; its "s ago" counter starts when this state appears, so it understates the age.

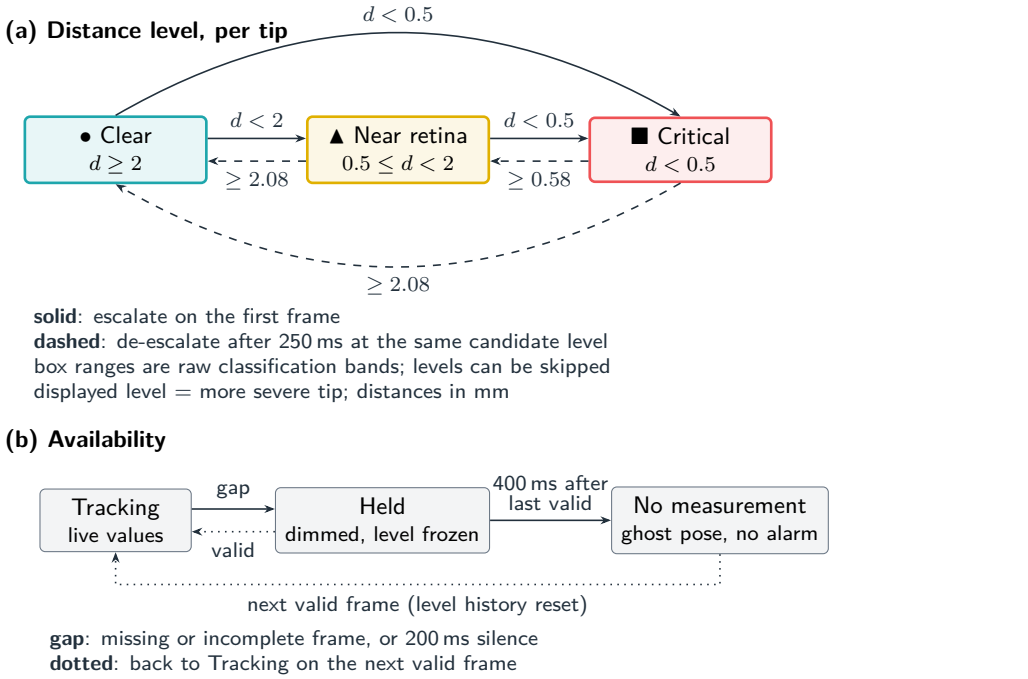


Figure 3: Two independent state axes. (a) Hysteresis and dwell act only on the way down. (b) Missing data is never read as safe: it holds the level for at most 400 ms after the last valid measurement and then removes it. Not drawn: *Stale* (source timestamp frozen for 1.5 s) and *Disconnected* (socket error, no grace).

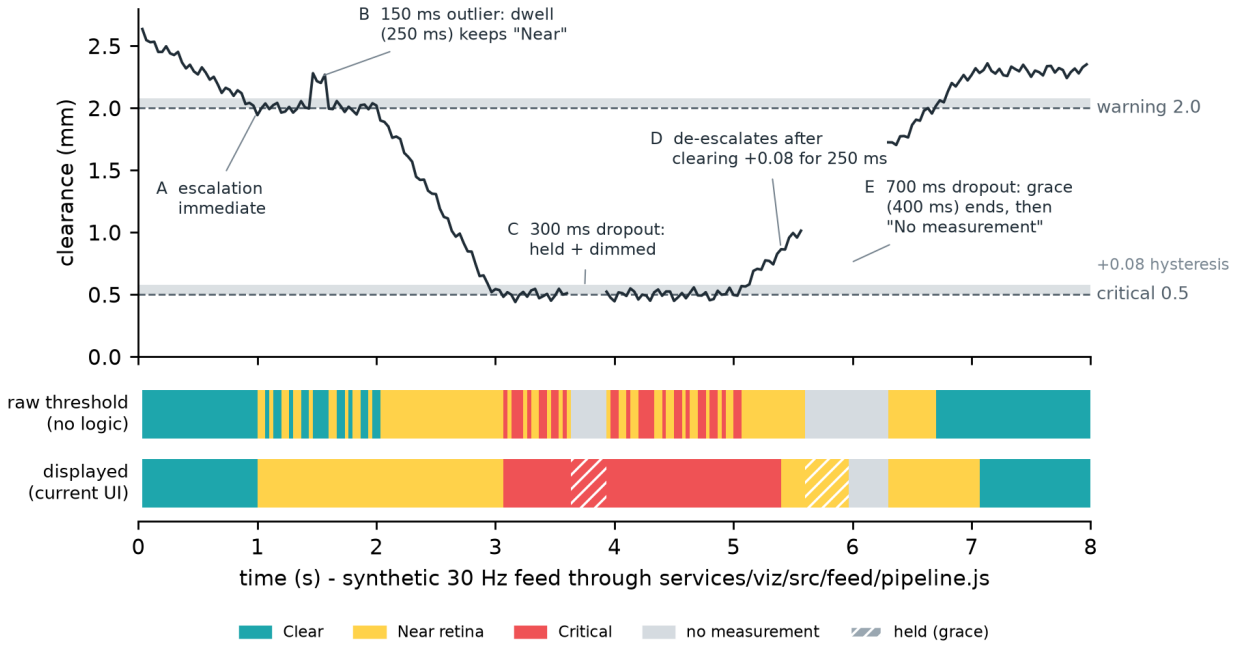


Figure 4: The pipeline code of the workstation processing a synthetic 30 Hz feed. Top: input clearance with the thresholds and the 0.08 mm hysteresis bands. Bottom: the raw threshold classification flickers; the displayed level escalates immediately (A), ignores a 150 ms outlier (B), holds through a 300 ms dropout (C), releases after 250 ms above the band (D) and, in a 700 ms dropout, shows "No measurement" once the 400 ms grace has expired (E).

Table 1: Replay of the three simulator runs, emulating the rules on the reconstructed tip midpoint. Previous: 0.05 mm hysteresis, no dwell, no grace; current: the rules of Figure 3. The replay through the actual UI pipeline (both tips, minimum shown) gives 54.9 s of false critical time.

metric (pooled over 381 s)	previous	current
status changes per minute (incl. availability)	106.5	37.8
Near/Critical episodes ended by a dropout	126	39
missed critical time (s)	0.63	0
false critical time (s)	13.8	43.4
time shown as "No measurement" (%)[a]	26.5	15.6

[a] held values during the grace period count as shown.

We replayed all three simulator recordings (11 428 frames, 381 s) through a midpoint emulation of the previous and the current rules (the previous rules no longer exist in code) and compared the result with the logged midpoint clearance, our ground-truth proxy (Table 1). The new rules cut status changes by almost two thirds and, by holding through its dropouts, recover the 0.63 s of the one truly critical episode that the previous rules missed (one episode, so no evidence of detection accuracy). The cost is specificity: holding and slower release make episodes longer, and most episodes are false to begin with. Because the tracker mostly places the tip *closer* than it is (80 % of valid reconstructions, median −0.53 mm), the estimate often drops below 0.5 mm while the tip is still millimetres away, and Critical then stays until the release rules are met. A second replay feeds both reconstructed tips through the UI pipeline itself, with the same proxy. In it, the logged clearance at the onset of a Critical display was 3.0 mm (median), and only 3.2 % of the time Critical was shown had a logged clearance below 0.5 mm. We draw two consequences in the design. First, the live Critical frame is steady red rather than the flashing red that IEC 60601-1-8 [3] associates with high-priority alarms, so that a mostly false red signal does not teach trainees to ignore red [1]. There is also no audible alarm; this was a project decision, and alarm fatigue is a further reason for it. Second, the accuracy note states the problem openly instead of hiding it. The UI does not correct the bias. It belongs to the orthographic camera model; on one recorded run, a perspective prototype with a camera fitted to that session reduced the median height bias from −0.50 to −0.11 mm.

1.3 Data path and limitations

The workstation does not average or filter the numeric clearances; levels change only by the rules above, and the approach rate is smoothed (upstream, the live detector smooths its 2D points with an exponential moving average, weight 0.5). The readout refreshes at most every 100 ms, but a change of level or availability is shown at once. Live instrument poses are interpolated between two measured poses (about 80 ms behind, never extrapolated) to remove stutter in the magnified scope. At the median approach speed within the last 3 mm in the recordings (0.53 mm/s), this interpolation lag corresponds to 0.04 mm, small compared with the measurement error.

The design has been checked against recorded data, not against users. Three runs by one user contain a single truly critical episode. The thresholds are provisional, and hysteresis, dwell and grace were derived and evaluated on the same data. Although the grace period removed most dropout-ended Near/Critical episodes in the midpoint emulation, data loss still ends most Critical episodes in the UI replay (32 of 58 observed endings followed grace expiry). In live operation, only reconstruction failures arrive as gaps: after a detection dropout or a detector stall, the stream server re-sends the last pose with a fresh send time, so the screen shows a frozen but current-looking value. A capture timestamp from the tracker would fix this. The recordings log only the tip midpoint, so the accuracy of Q2 (which tip is closer) is unmeasured. The 3D position error (median 1.4–2.5 mm) is larger than the height error, so the reticle and the scope can look precise while being misplaced laterally. The next steps are a heuristic evaluation and a small think-aloud study with trainees, and a replay of new recordings once the perspective model is in place.

References

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